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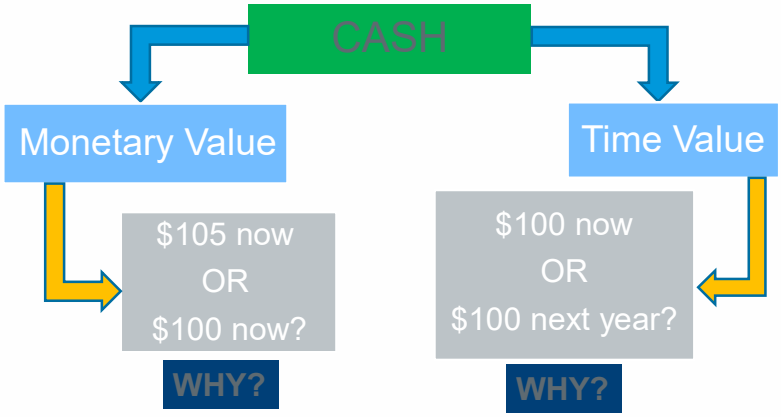
F78AB
Actuarial and Financial Mathematics

Chapter 2: Interest and Present Value

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Properties of Cash



$100 now OR $105 next year?

How to decide?

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$1,000,000 now or $1,000 a week for the rest of your life - which would you choose?

20-year-old lottery winner decides against $1M lump sum, opts for lifetime weekly annuity in hopes of buying a home

This Gen Z winner turned down $1M cash for steady income after winning big with Gagnant à Vie

Leticia Gaba, Writer, Yahoo News Canada
Mon, September 15, 2025 at 1:00 p.m. GMT+1
2 min read



Brenda Aubin-Vega's win is one of ten annuity prizes claimed in Québec this year. (Courtesy: Loto-Québec)

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2.1 The Time Value of Money

- Would you rather have £100 now or next week? Why?
  - **Deferred use** – you do not have use of the money
  - **Risk** – the money might never be paid to you
  - **Inflation** – prices might rise

So, you expect a reward for letting someone else have use of your money for a period of time – this is known as **interest.**

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2.1 The Time Value of Money

- The amount of money deposited or invested is known as **capital** or **principal**.

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E.g. 2.1.1: Interest rate calculation

- You deposit £100 in the savings account of Bank ABC for one year.
- In one year's time you **receive £105**.
- This £105 consists of the return of your **capital** (£100) and **interest** (£5).
- You can deduce that the bank paid you **interest at the rate of 5% per annum**.

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E.g. 2.1.2: Accumulated Value

- You deposit £100 in the savings account of Bank XYZ for one year.
- The bank pays interest at 3% per annum.
- In one year's time you will **receive £103**, which consists of the return of your **capital** (£100) and **interest** (£3).

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2.2 Effective Rate of Interest

- The **effective rate of interest per unit of time** is the amount of interest received at the end of one time unit per £1 (or other currency) invested at the start of that time unit.
- The effective rate of interest per time unit is most often denoted i.
- Other letters used are j, or subscripted symbols like i_1 and i_2.

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Simple vs Compound Interest

- We know that £1 invested for one time unit at the effective interest rate of i per time unit becomes $(1+i)$.
- How much will you have if you invested £1 for two, or more, time units at an effective rate of interest i per time unit?
- The answer depends on how the interest is calculated.

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Simple vs Compound Interest

- **Simple interest**
  - Interest is earned only on the initial capital deposited. **Interest does not earn interest.**
  - Suppose an amount £1 is deposited into an account that pays **simple interest** at the rate of *i* per time unit.

    At the end of *n* time units the deposit would become: **£ (1+*ni*)**

Give a numerical example of this to make easier to visualise, draw the interest payments on a timeline, eg 5% over 3 years – for £100 invested get £ 5 interest each year, total £115 after 3 years

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Simple vs Compound Interest

- **Compound interest**
  - Interest is earned on the initial capital deposited and from the interest that has been paid into the account. **Interest earns interest.**
  - Suppose an amount £1 is deposited into an account that pays **compound interest** at the rate of *i* per time unit. At the end of *n* time units the deposit would become **£ $(1+i)^n$**

Numerical example on board – 2% compound over 3 years for £100 invested. Interst in Y1 £5, Y2 is 5%x£105 = £5.25, Y3 is 5% of £110.25 = £5.52
Total after 3 years = 115.77 (= 100 x 1.05^3) with rounding error!

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Important

See week 1 additional exercise on Canvas to illustrate the difference between applying simple and compound interest

We will assume Compound Interest in this course unless otherwise stated.

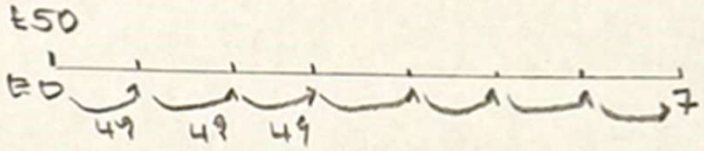
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2.3 More than One Time Unit

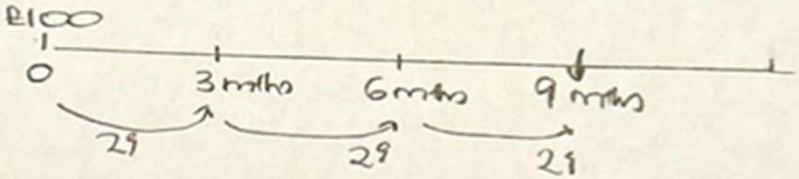
- Ex 2.3.1:
  - Invest £50 for 7 years at 4% per annum effective.
- Ex 2.3.2 :
  - Invest £100 for 9 months at 2% per quarter-year effective.
- Ex 2.3.3 :
  - Invest £80 for 1 year at 6.25% per half-year effective.

Round your answers to 2 decimal places as these are monetary amounts.

(1) $50 \times (1.04)^7 = £65.80$ (2 dp)



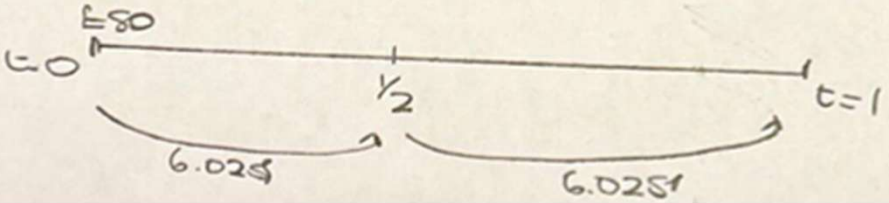
(2) $100 \times (1.02)^3 = £106.12$



(3) ~~$80 \times (1.04)^7 = £105.27$~~

$80 \times (1.0625)^2$

$= £90.31$



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2.4 Compounding over any number of time units

The compound interest formula $(1+i)^t$ extends to any real number $t \geq 0$.

- Ex 2.4.1
  - Invest £2,000 for 4 months at 10.5% per annum effective.
- Ex 2.4.2
  - Invest £2,500 for 1 month at 2% per quarter year effective.
- Ex 2.4.3
  - Invest £100 for 3.6 years at 4.1% per half year effective.

1. Effective interest rate per 4 months is $(1.105)^{4/12} - 1 = 3.338418\%$
Accumulated value = 2000 x 1.0228418 = 2,067.68

2. Effective interest rate per month is $(1.02)^{1/3} - 1 = 0.662271\%$
Accumulated value = 2500 x 1.00662271= 2516.56

3. Effective interest rate for 3.6 years is $(1.041)^{(3.6x2)} - 1 = 33.550418\%$
Accumluated value = 100 x 1.33550418 = 133.55

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2.5 Changing the Time Unit

- A contract may have interest rates and payments that have different time units.
- For example, a mortgage is a contract that has monthly cash flows but annual interest.
- It would be more convenient to work with an effective rate of interest per month.

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2.5 Changing the Time Unit – for example

- Say the effective rate of interest per year is i, and the effective rate of interest per month is j
- Based on an initial investment of £1, in one year's time we will have:
  £(1+*i*) based on the effective interest rate per year; and
- **£$(1+j)^{12}$** based on the effective interest rate per month

Draw time line to denote the interest payments at end of year and monthly amounts

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2.5 Changing the Time Unit

- The two effective rates of interest are defined to be equivalent if we have the same amount of money after one year.
- Hence after 1 year we have:

$$£(1 + i) = £(1 + j)^{12}$$

so $j = (1 + i)^{1/12} - 1$

We can use this approach to calculate the effective interest rate over any time period.

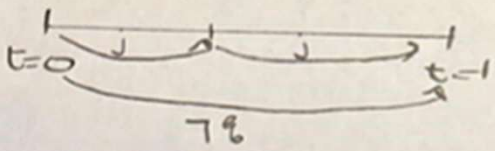
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2.5 Changing the Time Unit

- Ex 2.5.1
  - What effective rate of interest per half-year is equivalent to 7% per annum effective?
- Ex 2.5.2
  - What effective rate of interest per quarter-year is equivalent to 15% per two years effective?
- Ex 2.5.3
  - What effective rate of interest per 1.7 years is equivalent to 1.3% per month effective?

2.5 Changing the Time Unit

- Ex 2.5.1
  - What effective rate of interest per half-year is equivalent to 7% per annum effective?
- Ex 2.5.2
  - What effective rate of interest per quarter-year is equivalent to 15% per two years effective?
- Ex 2.5.3
  - What effective rate of interest per 1.7 years is equivalent to 1.3% per month effective?



(1) If j is the effective rate of interest per half-year:

$$(1+j)^2 = (1.07) \Rightarrow$$

$$\Rightarrow j = (1.07)^{1/2} - 1 = 0.0344080$$

or 3.4408 %.

(2) If j is the effective rate per quarter-year:

$$(1+j)^8 = (1.15)$$

$$\Rightarrow j = 0.01762374 \text{ or } 1.762374\%.$$

(3) $$(1+j) = (1.013)^{1.7\times 12}$$

$$\Rightarrow j = 0.301466 \text{ or } 30.1466\%.$$

1

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Comment on Rounding

Read "Comment on Rounding" in Chapter 2 of notes.

- Monetary amounts **must not** be shown to more than two decimal places.
- For interest rates our convention will be to calculate to at least six decimal places.

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2.6 Accumulating and Discounting

- We say that money accumulates with interest. The amount to which money accumulates (with the addition of interest) is its accumulation or accumulated value (AV or sometimes Future Value - FV).
- If i is the effective rate of interest per time unit, we call $(1+i)^t$ the accumulation factor over t time units.
- Money required **now** to meet a known future payment is called the discounted value or present value (PV) of that payment.

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Present values and accumulated values

- The Present Value (now) of a known future payment will always be less than that payment amount – because we assume we can invest money now, to earn interest and help pay the payment when it falls due.

- Think about how this looks on a timeline

Draw simple timeline to illustrate

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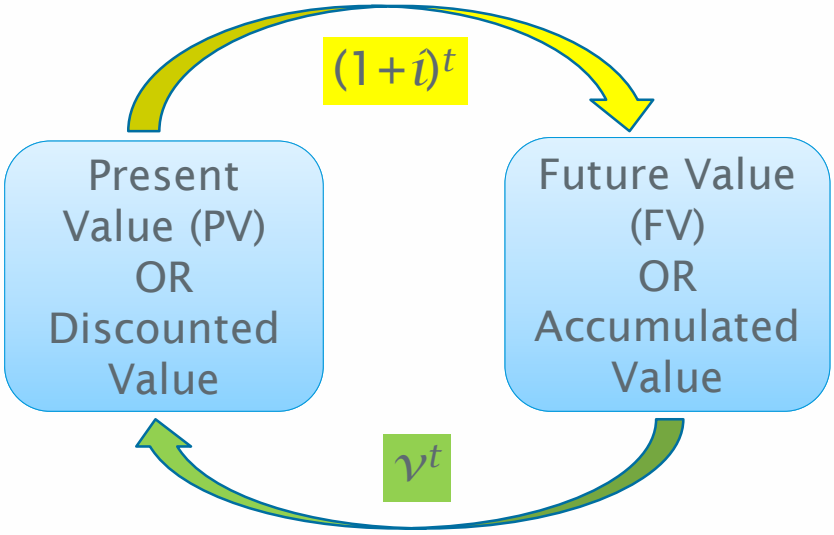
2.6 Accumulating and Discounting

- If i is the effective rate of interest per time unit, we call $(1+i)^{-1}$ the discount factor per time unit.
- Definition: $v = (1+i)^{-1}$

- We tend to write v^t for the discount factor over t time units.
- Accumulating and Discounting are inverse operations.

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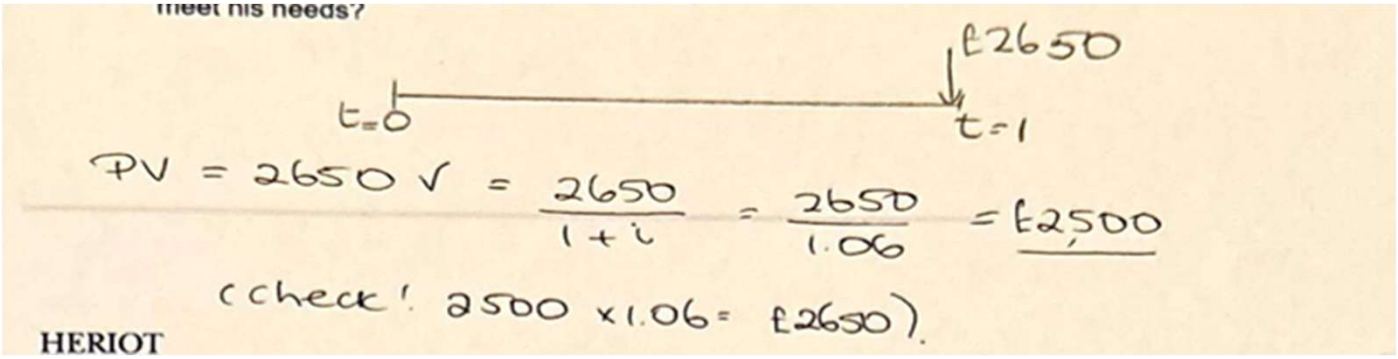
2.6 Accumulating and Discounting



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2.6 Accumulating and Discounting

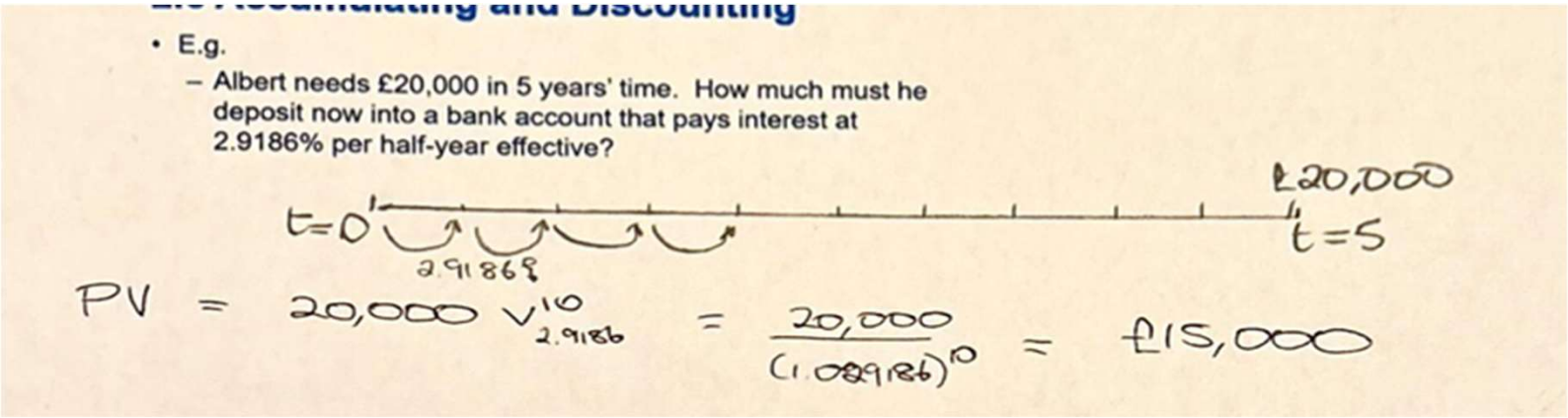
- E.g.
  - Joseph needs £2,650 in 1 year's time. The bank offers interest at 6% per annum. How much must he deposit now to meet his needs?



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2.6 Accumulating and Discounting

- E.g.
  - Albert needs £20,000 in 5 years' time. How much must he deposit now into a bank account that pays interest at 2.9186% per half-year effective?



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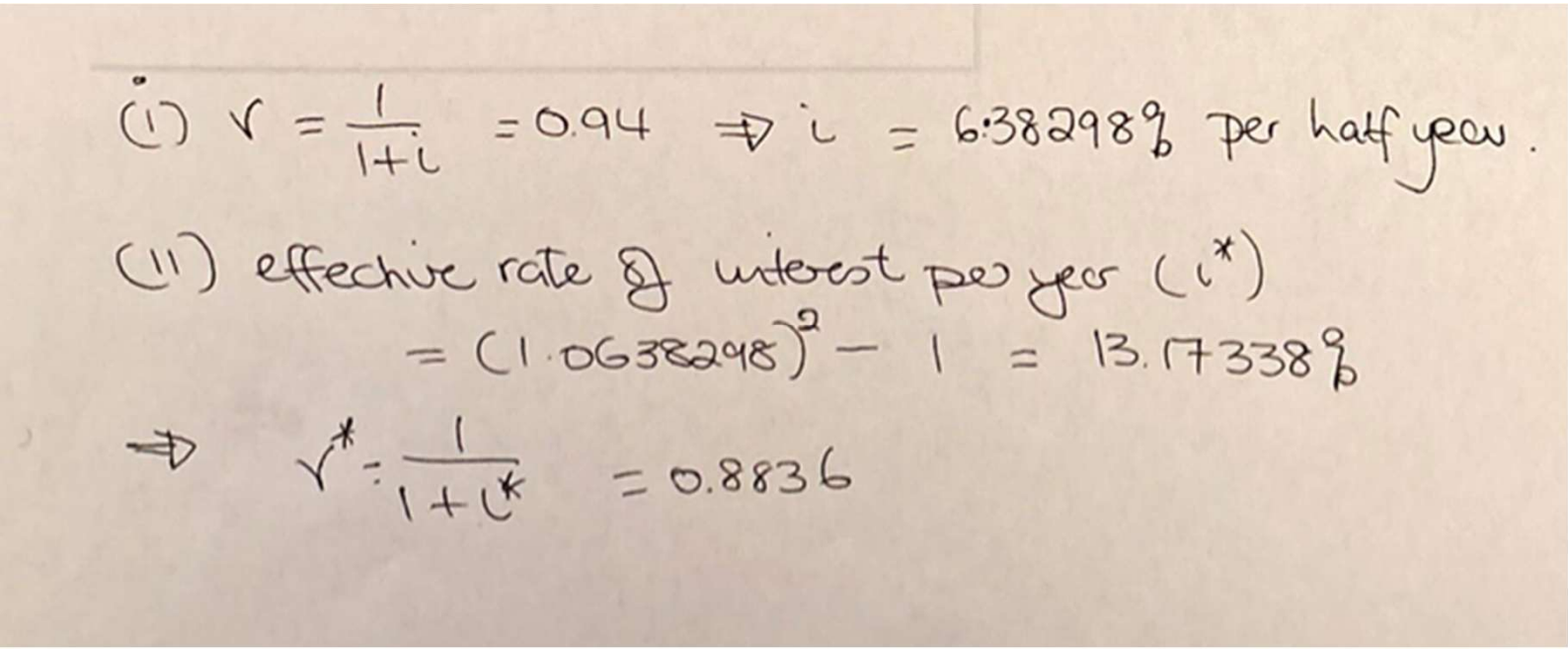
2.6 Accumulating and Discounting

- E.g.
  - Let $v = 0.94$ be the discount factor per half year.

    i) What is the effective rate of interest per half-year?

    ii) What is the equivalent discount factor per year?

    Give your answers to 6 decimal places.



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2.7 The Effective Rate of Discount

- **Definition:** Let i be the effective rate of interest per time unit, so that $v = (1+i)^{-1}$ is the corresponding discount factor.

  The effective rate of discount per unit of time, d, is defined as $d = 1 - v$.

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2.7 The Effective Rate of Discount cont..

- Hence, for each of the operations of accumulating and discounting, we have an **effective rate** and a corresponding **factor:**

Operation	Rate	Factor
Accumulation	i	$1+i$
Discounting	d	$1-d=v$

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Useful Results

	Value of		
In terms of	i	v	d
i	-	$\frac{1}{i+1}$	$\frac{i}{1+i}$
v	$\frac{1}{v}-1$	-	$1-v$
d	$\frac{d}{1-d}$	$1-d$	-

$$\boldsymbol{iv = d}$$

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2.7 The Effective Rate of Discount

- E.g.
  - Given $i = 0.04$ effective per time unit, find v and d.

– Given $i = 0.04$ effective per time unit, find v and d.

$$v = \frac{1}{1+i} = \frac{1}{1.04} = 0.961538$$

$$d = \frac{i}{1+i} = \frac{0.04}{1.04} = 0.0384615$$

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2.8 The Value at Time s of a Cashflow at Time t

Say we have a cashflow X at time t and that the effective rate of interest is i per time unit.

- **Case 1:** $s \geq t$, *ie we are looking forward*
  - We must accumulate the cashflow from time t to time s,

$$X(1+i)^{s-t}$$

-

Draw time lines to illustrate both case 1 and case 2 (next slide)

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2.8 The Value at Time s of a Cashflow at Time t, *continued*

For the same cashflow X at time t, with effective rate of interest i per time unit.

- **Case 2:** $s < t$, *ie we are looking backwards*
  - We must discount the cashflow from time t to time s,

$$X(1+i)^{-(t-s)} = Xv^{t-s}$$

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Remember!

- **Project (accumulate) forward!**

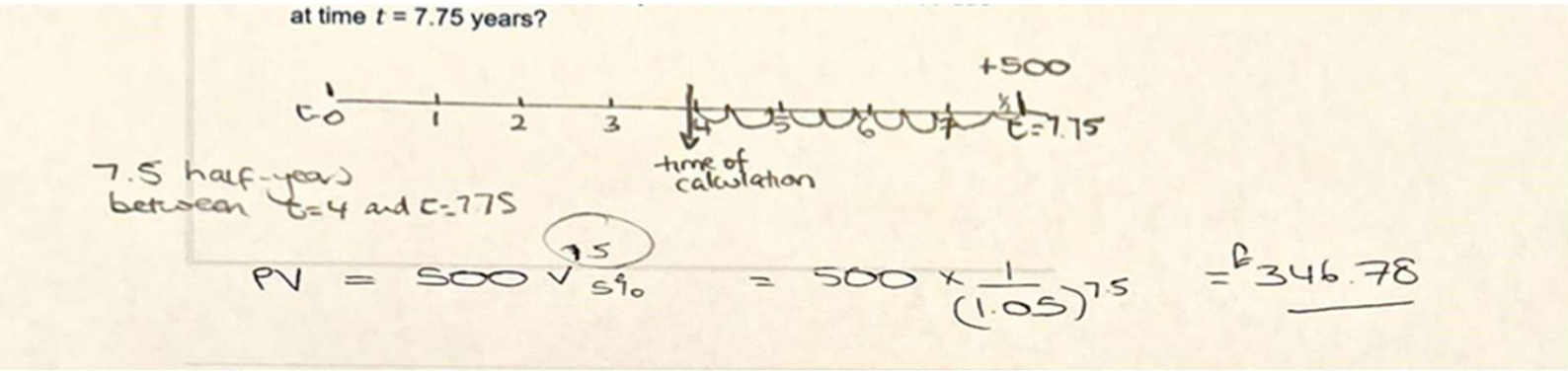
- **Discount backwards!**

  Use a timeline to ensure you understand the cashflows, interest rates and time periods

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E.g. Assume the effective rate of interest is 5% per half year.

What is the value at time s = 4 years of a cashflow of £500 due at time t = 7.75 years?



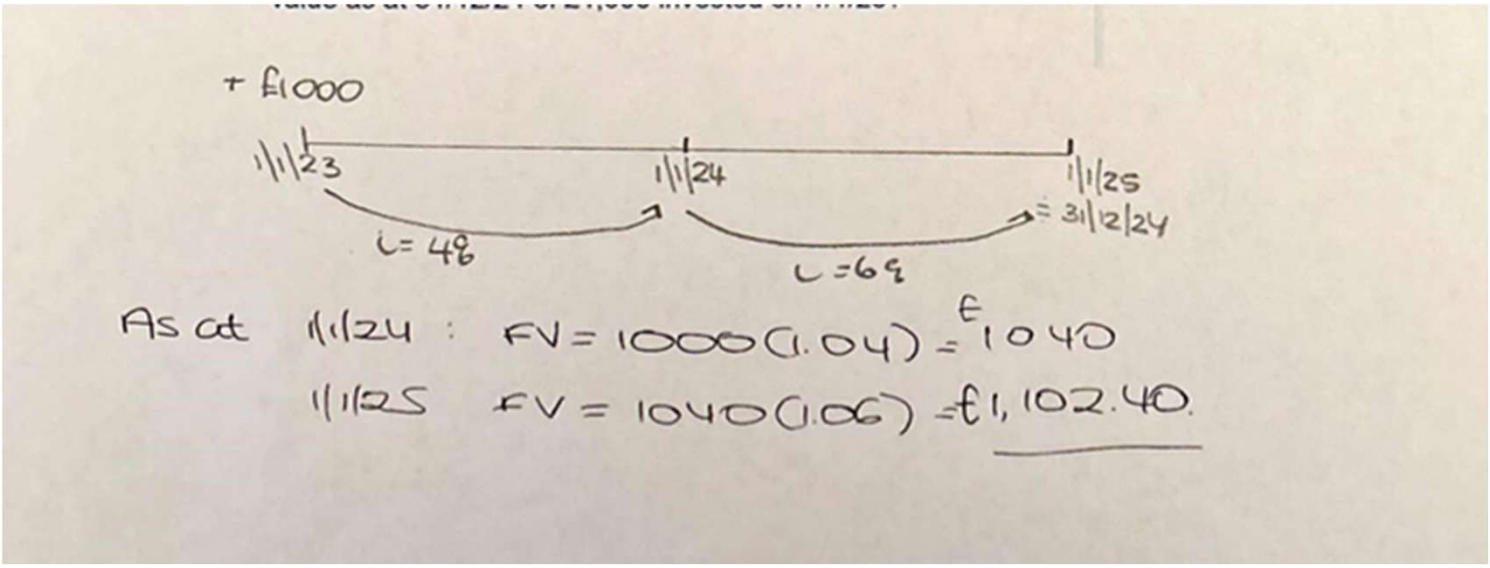
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2.9 Non-Constant Interest Rates

Interest rates are not always constant.

We can use the same theory as earlier to determine present values and accumulations if we know the interest rate over different periods of time.

- E.g.
  - Let the effective interest rate be 4% per annum during 2023 and 6% per annum during 2024. What is the accumulated value as at 31/12/24 of £1,000 invested on 1/1/23?



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2.9 Non-Constant Interest Rates

- E.g. 2.9.2

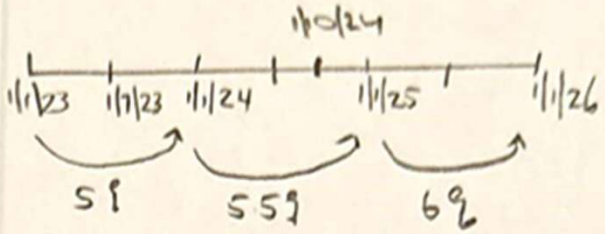
The effective rate of interest will be 5% per annum during 2023, 5.5% per annum during 2024 and 6% per annum during 2025. Calculate the accumulation of each of the following cashflows.

1. £1,000 invested on 1/1/23 for 3 years?
2. £1,000 invested on 1/7/23 for 2 years?
3. £1,000 invested on 1/10/24 for 1 year?

- E.g. 2.9.2

The effective rate of interest will be 5% per annum during 2023, 5.5% per annum during 2024 and 6% per annum during 2025. Calculate the accumulation of each of the following cashflows.

1. £1,000 invested on 1/1/23 for 3 years?
2. £1,000 invested on 1/7/23 for 2 years?
3. £1,000 invested on 1/10/24 for 1 year?



(1) $FV = 1000(1.05)(1.055)(1.06) = 1,174.22$

(2) $FV = 1000(1.05)^{1/2}(1.055)(1.06)^{1/2} = 1,113.01$

(3) $FV = 1000(1.055)^{1/4}(1.06)^{3/4} = 1,058.75$

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2.9 Non-Constant Interest Rates

In general suppose that an amount of X is invested between time t_o and t_n with effective interest rate being i_r between times t_{r-1} and t_r then the accumulated value of X at time t_n is

$$X\ (1+i_1)^{t_1-t_0}\ (1+i_2)^{t_2-t_1}\ \ldots\ (1+i_n)^{t_n-t_{n-1}}$$

$$= X \prod_{r=1}^{n} (1+i_r)^{t_r-t_{r-1}}$$

Can draw a timeline to illustrate this in general case

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2.9 Non-Constant Interest Rates

Alternatively suppose we require the Present Value (PV) at time t_o of *X* due at time t_n. Then this PV is

$$X\ (1+i_1)^{t_0-t_1}\ (1+i_2)^{t_1-t_2}\ \ldots\ (1+i_n)^{t_{n-1}-t_n}$$

$$= X \prod_{r=1}^{n} (1+i_r)^{t_{r-1}-t_r}$$

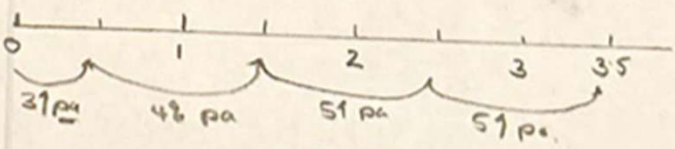
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2.9 Non-Constant Interest Rates

- E.g.2.9.3

Starting now, the effective rate of interest will be 3% per annum for 6 months, 4% per annum for the following one year, and 5% per annum for the following 2 years. What is the accumulation factor over the period [0, 3.5] (measuring time in years)?

Starting now, the effective rate of interest will be 3% per annum for 6 months, 4% per annum for the following one year, and 5% per annum for the following 2 years. What is the accumulation factor over the period [0, 3.5] (measuring time in years)?



Acc factor $= (1.03)^{1/2}(1.04)(1.05)^2$

$= 1.163672$

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2.9 Non-Constant Interest Rates

- E.g. 2.9.4

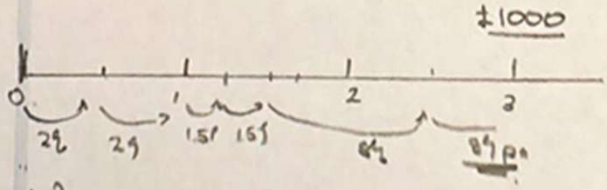
Starting now, the effective rate of interest will be 2% per half year for 1 year, 1.5% per quarter year for 6 months, and 8% p.a. for the following 18 months. What is the PV now of £1,000 due in 3 years' time?

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2.9 Non-Constant Interest Rates

• E.g. 2.9.4

Starting now, the effective rate of interest will be 2% per half year for 1 year, 1.5% per quarter year for 6 months, and 8% p.a. for the following 18 months. What is the PV now of £1,000 due in 3 years' time?



calc^ as at t = 0

⇒ discounting back

$$PV = 1000\left(1.08\right)^{-\frac{1}{2}}\left(1.08\right)^{-1}\left(1.015\right)^{-3}\left(1.02\right)^{-2}$$

or

$$PV = \frac{1000}{(1.08)^{\frac{1}{2}}(1.08)(1.015)^{3}(1.02)^{2}}$$

$$= \frac{1000}{1.203007} = £831.25$$

4

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2.10 General Accumulation and Discount Factors

It is sometimes convenient to work with a more general and abstract model of accumulation and discounting.

Definition

For arbitrary times t and s, with $t \leq s$, define

- $V(t, s)$ = present value at time t of 1 payable at time s.
- $V(s, t) = V(t, s)^{-1}$ = accumulation at time s of 1 payable at time t.

Draw timeline to illustrate general case

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2.10 General Accumulation and Discount Factors

Rules

- $V(t, s) > 0$ for any t and s.
- $V(t, s) = V(t, r)V(r, s)$ for any t, s and r.

Question

Show that $V(t, t) = 1$.

Solution is in the course notes

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Next chapter...

Chapter 3: Cashflows and Annuities

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